

Formulation and In Vitro-In Vivo Evaluation of Black Raspberry Extract-Loaded PLGA/PLA Injectable Millicylindrical Implants for Sustained Delivery of Chemopreventive Anthocyanins



Purpose The objective of this study was to formulate and evaluate freeze-dried black raspberry (FBR) ethanol extract (RE) loaded poly(DL-lactic-co-glycolic acid) (PLGA) and poly(DL-lactic acid) (PLA) injectable millicylindrical implants for sustained delivery of chemopreventive FBR anthocyanins (cyanidin-3-sambubioside (CS), cyanidin-3-glucoside (CG) and cyanidin-3-rutinoside (CR)). **Methods** Identification and quantitation of CS, CG, and CR in RE was performed by mass spectroscopy and HPLC. RE:triacetyl- β -cyclodextrin (TA- β -CD) inclusion complex (IC) was prepared by a kneading method and characterized by X-ray diffraction (XRD), nuclear magnetic resonance spectroscopy (NMR) and UV-visible spectroscopy. RE or RE:TA- β -CD IC-loaded PLGA or PLA implants were prepared by a solvent extrusion method. In vitro and in vivo controlled release studies were conducted in phosphate-buffered saline Tween-80 (pH 7.4, 37°C) and after subcutaneous administration in male Sprague-Dawley rats, respectively. Anthocyanins were quantified by HPLC at 520 nm. **Results** The content of CS, CG, and CR in RE was 0.2, 1.5, and 3.5 wt%, respectively. The chemical stability of anthocyanins in solution was determined to be pH-dependent, and their degradation rate increased with an increase in pH from 2.4 to 7.4. PLGA/PLA millicylindrical implants loaded with 5 or 10 wt% RE exhibited a high initial burst and short release duration of anthocyanins (35–52 and 80–100% CG + CR release after 1 and 14 days, respectively). The cause for rapid anthocyanins release was linked to higher polymer water uptake and porosity associated with the high osmolytic components of large non-anthocyanin fraction of RE. XRD, ¹H NMR and UV-visible spectroscopy indicated that the non-anthocyanin fraction molecules of RE formed an IC with TA- β -CD, decreasing the hydrophilicity of RE. Formation of an IC with hydrophobic carrier, TA- β -CD, provided better in vitro/in vivo sustained release of FBR anthocyanins (16–24 and 97–99% CG + CR release, respectively, after 1 and 28 days from 20 wt% RE:TA- β -CD IC/PLA implants) over 1 month, owing to reduced polymer water uptake and porosity. **Conclusion** PLA injectable millicylindrical implants loaded with RE:TA- β -CD IC are optimal dosage forms for 1-month slow and continuous delivery of chemopreventive FBR anthocyanins.